

COURSE DESCRIPTION			
Program	Food Processing Technology		
Course Title	Food Processing Technology		
Course Code	2421	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

Food Processing Technology deals with the conversion of raw agricultural materials into safe, nutritious, and convenient food products. The course covers processing methods, preservation techniques, equipment used in food industries, and quality control. Its main aim is to reduce post-harvest losses, improve shelf life, ensure food safety, and add value to foods while maintaining nutritional and sensory quality.

Course Objectives

1	To understand the various cleaning techniques for raw materials before processing
2	To learn the concepts of drying, freezing and chilling in raw materials
3	To impart knowledge on extrusion and related concepts

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge of different components of food	1421	Fundamentals of food processing technology	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the concept of raw material preparation by high temperature processing	Understand
CO2	Discuss size reduction principles and equipment for solid foods	Understand
CO3	Illustrate the importance of drying, refrigeration and freezing during processing	Understand
CO4	Summarize the concepts of extrusion and minimal processing techniques in food technology sector.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	3	-	-	-	-	-	-	-	-	-	-
Correlation Level	3	-	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total
4	0	0	0	6	60	4	40	60	100	0	0	0

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Raw material preparation and thermal processing in food industries	cleaning techniques,pasteurization&blanching	13	0
Module II	Size reduction and homogenization in food processing	principles of size reduction, equipment used for size reduction,homogenization in food processing	13	0
Module III	Summarize the concepts of extrusion and minimal processing techniques in food technology sector.	extrusion process,minimal processing of fruits and vegetables ,hurdle technology	17	0
Module IV	Illustrate the importance of drying, refrigeration and freezing during processing	drying,refrigeration ,chilling& freezing	17	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Various methods of cleaning of raw materials-dry and wet techniques	Explain raw material preparation steps including cleaning,sorting,grading	L	CO1	Understand	3
pasteurization-principle, process and types with its importance- LTLT, HTST, UHT	Describe the principles and methods of pasteurization	L	CO1	Understand	6
Principle ,types, process of blanching. and its application	Analyze effects of heat treatment on food quality	L	CO1	Understand	4
Module II					
Equipments used in size reduction- roller mill, impact mill, attrition mill	classify types of equipments used for size reduction	L	CO2	Understand	6
principle and laws of size reduction - Kicks, Bond, Rittinger's law	Explain the difference and application of these laws in size reduction	L	CO2	Understand	4
Importance and principle of Homogenization	explain how homogenization improves stability and quality of food products	L	CO2	Understand	3
Module III					
Extrusion - Theory, operating characteristics & application	Explain the principle & theory of extrusion processing	L	CO3	Understand	4
different methods of extrusion and its equipments	Describe the working of extruders	L	CO3	Understand	5
Principle, Process and applications of various minimal processing techniques- Ohmic heating, RF heating, high pressure processing	Explain the principles and applications of minimal processing techniques such as ohmic heating, RF heating, and high-pressure processing	L	CO3	Understand	4
Hurdle technology - Principle and application.	Describe the concept and uses of hurdle technology in food processing	L	CO3	Understand	4
Module IV					
Driers used in food processing industry- Tray, tunnel, fluidized bed, spray, Rotary drier	Identify types of driers used and their mechanism	L	CO4	Understand	6
VA,VC refrigeration systems- components - compressor, condenser, evaporator, refrigerant, COP	Describe the two types of refrigeration system	L	CO4	Understand	5
mechanism of Chilling and freezing and their difference	Differentiate between chilling and freezing	L	CO4	Understand	3
types of freezers and their application ,effects of freezing in food	Explain he types of freezers in food industries	L	CO4	Understand	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		

1	Think-Pair-Share Activity Ask students: "Why is blanching done before freezing vegetables?"Think individually (2 minutes) Discuss with a partner (3 minutes),Share answers with class	6
2	Process Sequencing Activity - Provide jumbled steps of pasteurization / sterilization.Arrange steps in correct order and explain each step.	12
Module II		
5	Equipment-Principle Matching Provide two columns: • Column A: Hammer mill, Roller mill, Cutter, Ball mill • Column B: Impact, Compression, Shear, Attrition Match equipment with working principle and justify choice	12
6	Object-Based Analogy Activity - Show pictures / real items (nuts, grains, flour). Ask students which equipment is suitable and why.	12
Module III		
9	Comparative Group Chart Divide class into 3 groups: • Group 1: Drying • Group 2: Refrigeration • Group 3: Freezing Each group prepares a chart with: • Temperature range • Shelf life • Advantages • Examples	12
10	Real-Life Product Identification - Ask students to list food products they use daily that are:Dried,Refrigerated,Frozen Explain why each method is used.	12
Module IV		
13	Video-Based Discussion - Show a short video of an extruder or MAP packaging.	12
14	Role-Play Activity - Students act as:Foodtechnologist,Consumer,Manufacturer. Choosing between minimally processed salad and conventional processed food.	12
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Food Processing Technology: Principles and Practice	P.J. Fellows	
2	Food Science and Technology	Bhupendra Singh Khatkar	

REFERENCE			
Sl. No.	Title	Author	Publication
1	Introduction to Food Engineering	R. Paul Singh & Dennis R. Heldman	
2	Food Processing Handbook	James G. Brennan & Alistair S. Grandison	

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.ajce.in/fe/downloads/FoodTechnologysyllabus%28s5%2Cs6%29.pdf?utm_source=chatgpt.com	I & II
2	https://agriculture.institute/food-fundamentals-cpo/unit-operations-food-processing-techniques-applications/?utm_source=chatgpt.com	III & IV

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60

Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

SITTTR